

Discrete Mathematics — Homework 3: Inference and First Order Logic (FOL)

Solutions

1. Validity

Question being asked. Consider the inference rule: $(p \rightarrow q), q \Rightarrow p$, where p = “I am cold” and q = “I forgot my jacket”. Determine whether it is valid, explain briefly, and verify via a truth table whether $((p \rightarrow q) \wedge q) \rightarrow p$ is a tautology.

Answer. The rule is **not valid**; it is the fallacy of *affirming the consequent*. A counterexample is $p = F$ and $q = T$: then $p \rightarrow q$ is true and q is true, but p is false.

6pt]

p	q	$(p \rightarrow q)$	$(p \rightarrow q) \wedge q$	$((p \rightarrow q) \wedge q) \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

Since the final column is false when $p = F$ and $q = T$, the formula is not a tautology, confirming invalidity.

2. Inference

Question being asked. Given premises:

- 1) $rainy \rightarrow (cold \vee windy)$
- 2) $\neg cold$
- 3) $windy \rightarrow uncomfortable$
- 4) $\neg uncomfortable$
- 5) $stormy \rightarrow rainy$

Prove $\neg stormy$, citing each rule/law.

Answer (derivation).

1. $rainy \rightarrow (cold \vee windy)$	Premise
2. $\neg cold$	Premise
3. $windy \rightarrow uncomfortable$	Premise
4. $\neg uncomfortable$	Premise
5. $stormy \rightarrow rainy$	Premise
6. Assume $stormy$	Assumption for contradiction
7. $rainy$	5, 6 Modus Ponens
8. $cold \vee windy$	1, 7 Modus Ponens
9. $windy$	8, 2 Disjunctive Syllogism
10. $uncomfortable$	3, 9 Modus Ponens
11. $\perp$	10, 4 Contradiction
12. $\neg stormy$	6–11 Negation Introduction (Indirect Proof)

3. FOL Propositions (Domain: Books)

Question being asked. Let the domain be $B = \text{“Books”}$. Translate into FOL using only the predicates $novel(x)$, $bestselling(x)$, $engaging(x)$, $wellwritten(x)$, $textbook(x)$, $difficult(x)$, $biography(x)$, $informative(x)$.

Answer.

- (1) $\forall x ((novel(x) \wedge bestselling(x)) \rightarrow (engaging(x) \wedge wellwritten(x)))$.
- (2) $\exists x (textbook(x) \wedge \neg difficult(x))$.
- (3) $(\forall x (biography(x) \rightarrow informative(x))) \wedge (\exists x (novel(x) \wedge bestselling(x)))$.

4. FOL Propositions (Domains: People and Movies)

Question being asked. Let the domain be $P = \text{“People”}$ and $M = \text{“Movies”}$. Translate (a)–(e) into FOL using predicates $comedy(y)$, $drama(y)$, $enjoys(x, y)$, $directed(x, y)$, $acclaimed(y)$.

Answer.

- (a) $\forall x \exists y (comedy(y) \wedge enjoys(x, y))$.
- (b) $\exists y (drama(y) \wedge \forall x enjoys(x, y))$.
- (c) $\forall x \exists y (comedy(y) \wedge \neg enjoys(x, y))$.
- (d) $\forall x \forall y (directed(x, y) \rightarrow enjoys(x, y))$.
- (e) $\forall y (acclaimed(y) \rightarrow \exists x directed(x, y))$.

5. Negation of FOL Propositions

Question being asked. For each formula, write an equivalent formula by pushing negations inward until each $\neg$ appears directly before a predicate.

Answer.

$$(a) \neg(\forall x (P(x) \rightarrow Q(x))) \equiv \exists x \neg(P(x) \rightarrow Q(x)) \equiv \exists x (P(x) \wedge \neg Q(x)).$$

$$(b) \neg(\exists x (R(x) \wedge \forall y S(x, y))) \equiv \forall x \neg(R(x) \wedge \forall y S(x, y)) \\ \equiv \forall x (\neg R(x) \vee \neg \forall y S(x, y)) \equiv \forall x (\neg R(x) \vee \exists y \neg S(x, y)).$$

$$(c) \neg(\forall x (\exists y P(x, y) \rightarrow \forall z Q(z))) \\ \equiv \exists x \neg(\exists y P(x, y) \rightarrow \forall z Q(z)) \\ \equiv \exists x (\exists y P(x, y) \wedge \neg \forall z Q(z)) \equiv \exists x (\exists y P(x, y) \wedge \exists z \neg Q(z)).$$